

AMMAAR SAADAT

✉ saadatam@umich.edu 🌐 AmmaarS.com 📞 734-620-4250 📄 linkedin.com/in/ammaar-saadat-0867a822a

Education & Awards

University of Michigan - Ann Arbor

Bachelor of Engineering - Computer Science and Engineering
Dean's Honors List

May 2025

Ann Arbor, Michigan

Michigander Scholars in Semiconductors Scholarship

Jan. 2025

Selected for a \$5,000 merit scholarship following semiconductor internship with KLA

Ann Arbor, Michigan

Courses: Operating Systems, Linear Algebra, Web Systems, Computer Architecture, UI Development, Computer Security, IR Systems, Data Structures and Algorithms, Calc I-III

Experience

KLA

May - Aug. 2024

Software Engineer - Infrastructure Intern — gRPC, RESTful APIs, Protobuf, Java, SpringBoot

Ann Arbor, Michigan

- Developed a semiconductor inspection infrastructure for standardized tool access by integrating gRPC, SpringBoot, and protobuf while collaborating with international engineering teams, successfully reducing infrastructure latency by 85%
- Engineered a SpringBoot microservice featuring 8+ low-latency RPC service endpoints by updating legacy distributed systems code

Sarf

Aug. - Dec. 2025

Founding Software Engineer — TypeScript.js, Next.js, React, SQL, iOS, Android

New York City, New York

- Led and designed end-to-end development of the scalable full-stack platform using TypeScript, SQL data models, live market-data APIs, and client/server state synchronization to enable user flows, raising up to \$200,000 within 5 months
- Implemented client-facing features and backend logic including user authentication, AI context workflows, and contract agreement pipelines to ensure user fulfillment

University of Michigan - Ann Arbor

Nov. 2023 - May 2024

Web Developer — WordPress, AwesomeTables, HTML5/CSS5, WPBakery

Ann Arbor, Michigan

- Iteratively engineered web pages representing guest speakers and esteemed professors for upcoming educational seminars

Projects

Vitra.AI Health | Software Engineer | Gemini, AI, Python, Flask, React, Project Management

May 2025

- Founded a real-time AI-powered speech translation platform supporting 40+ languages using Gemini, enabling low-latency streaming speech-to-text and multilingual output for clinical communication in production environments
- Optimized inference performance and translation accuracy through deployments with 2 clinics, improving latency and user experience in customer-facing production environments while collaborating with 4 engineers

Cat-thusiast - AI-Agent | Pinecone, Ollama, HuggingFace, AI, RAG, Python

March 2025

- Developed an conversational AI-Agent in Python that communicates cat facts via retrieval augmented generation (RAG) by comparing cosine similarities of vector embeddings of a data corpus
- Utilized Ollama-3 / HuggingFace inference models from a vector database via Pinecone

Raspberry Pi 5 Video Streaming System | Python, FastAPI, React, Next.js

Present

- Built a video streaming pipeline on a Raspberry Pi 5 using FastAPI and Picamera2, supporting live frame capture, JPEG encoding, and low-latency delivery over HTTP to a web-based client
- Automated full-stack deployment using Bash scripts for process orchestration, logging, dependency management, and system initialization across a local network environment

Multi-Threaded Network File Server | Operating Systems, Networking, C++, Threads

April 2025

- Designed a multi-threaded TCP file server in C++ in collaboration with 2 engineers, using thread-per-request concurrency, per-file locking, and hand-over-hand traversal to support 1,000+ concurrent client requests
- Ensured fault tolerance of 1,000+ concurrent client commands by implementing sanitization of malformed input, preventing race conditions with synchronization, and enabling disk crash consistency, verified through stress tests and I/O failure edge-case analysis

Technical Skills and Interests

Frameworks & Backend: TypeScript.js, Next.js, SQL, Flask, SpringBoot, Microservices, REST APIs

AI / APIs: OpenAI, Gemini, Pinecone, Ollama-3, gRPC

Languages: Python, JavaScript, C++, Java (SpringBoot), C, Bash, HTML/CSS5

Interests: Raspberry Pi, Volleyball, Marketing, Hiking, Gardening, Pokemon Go